ANALYZAONE

Analysis of Liver Dataset on the basis of Age, Gender and Hybrid

A PROJECT REPORT

Submitted by

Vani Varada 24MIP10023
Prathamesh Kumar 24MIP10034
Pushkar Singh 24MIP10083
Ritesh Kumar Jha 24MIP10088

in partial fulfilment for the award of the degree
of

INTEGRATED MASTER OF TECHNOLOGY

in

COMPUTER SCIENCE AND ENGINEERING
(COMPUTATIONAL AND DATA SCIENCE)

SCHOOL OF COMPUTING SCIENCE ENGINEERING AND ARTIFICIAL INTELLIGENCE (SCAI)

VIT BHOPAL UNIVERSITY

KOTHRIKALAN, SEHORE, MADHYA PRADESH - 466114

MARCH 2026

VIT BHOPAL UNIVERSITY, KOTHRIKALAN, SEHORE
MADHYA PRADESH – 466114

BONAFIDE CERTIFICATE

Certified that this project report titled **"ANALYZA-ONE"** is the bonafide work of **Vani Varada 24MIP10023, Pushkar Singh 24MIP10083, Prathamesh Kumar 24MIP10034, Ritesh Kumar Jha 24MIP10088** who carried out the project work under my supervision. Certified further that to the best of my knowledge the work reported at this time does not form part of any other project/research work based on which a degree or award was conferred on an earlier occasion on this or any other candidate.

Reviewer 1

Signature: _____
Name: _____
Designation: _____
Date: _____

Reviewer 2

Signature: _____
Name: _____
Designation: _____
Date: _____

Supervisor

Signature: _____
Name: _____
Designation: _____
Date: _____

Program Chair

Signature: _____
Name: _____
Designation: _____
Date: _____

The Project Exhibition is held on _____

ACKNOWLEDGEMENT

I am deeply indebted to **Dr. G. Viswanathan**, Chancellor of VIT Bhopal University, and **Dr. T.B. Sridharan**, Pro Vice-Chancellor, VIT Bhopal University, for their unwavering support and guidance. Their leadership and vision have been a constant source of inspiration throughout my academic journey.

I express my profound gratitude to **Dr. Pon Harshavardhan**, **Dean SCAI**, for his encouragement and invaluable support at every stage, which significantly contributed to the successful completion of this project.

My heartfelt thanks go to **Dr. Priscilla Dinkar Moyya**, **Program Chair**, Integrated M. Tech Computer Science & Engineering (Computational and Data Science), and **Dr. Venkat Prasad Padhy**, **Division Head**, AI Division, for their timely assistance, guidance, and invaluable insights during this project.

A special note of appreciation is reserved for my supervisor, **Dr. Priscilla Dinkar Moyya**, whose constant support, patience, and expertise were instrumental in completing this project. Her guidance, encouragement, and clarity in addressing my queries made this journey both meaningful and enriching.

I would like to express my sincere gratitude to my reviewers for their valuable feedback and insightful reviews. Lastly, I extend my deepest gratitude to my parents and friends, whose unwavering encouragement and support have been my driving force throughout this research endeavour.

Vani Varada 24MIP10023
Prathmesh Kumar 24MIP10034
Pushkar Singh 24MIP10083
Ritesh Kumar Jha 24MIP10088

LIST OF ABBREVIATIONS

Abbreviation	Full Form
ILPD	Indian Liver Patient Dataset
ML	Machine Learning
SMOTE	Synthetic Minority Over-sampling Technique
RF	Random Forest
SVM	Support Vector Machine
XGBoost	Extreme Gradient Boosting
LightGBM	Light Gradient Boosting Machine
LR	Logistic Regression
DT	Decision Tree
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
EDA	Exploratory Data Analysis
ALT / SGPT	Alanine Aminotransferase / Serum Glutamate Pyruvate Transaminase
AST / SGOT	Aspartate Aminotransferase / Serum Glutamate Oxaloacetate Transaminase
ALP	Alkaline Phosphatase
T-Bil	Total Bilirubin
D-Bil	Direct Bilirubin
NAFLD	Non-Alcoholic Fatty Liver Disease
TP	True Positive
TN	True Negative
FP	False Positive
FN	False Negative
CV	Cross-Validation
API	Application Programming Interface
UI	User Interface
UCI	University of California Irvine (ML Repository)

LIST OF FIGURES AND GRAPHS

Figure No.	Title	Page No.
Fig : 1	System Architecture Diagram of the ML Dashboard	23
Fig : 2	Data Processing Pipeline Flowchart	25
Fig : 3	Feature Importance - Random Forest vs XGBoost	27
Fig. 4.1	ROC Curves – All Models	28
Fig. 4.2	Histograms – Distribution of Important Features	29
Fig. 4.3	Boxplots – Features by Disease Class	29
Fig. 4.4	Confusion Matrices – All 6 Models (Full Dataset)	31
Fig. 4.5	Stacking Classifier Confusion Matrix	31
Fig. 4.6	Correlation Heatmap of all Features	32

LIST OF TABLES

Table No.	Title	Page No.
Table : 1	Pros And Cons Of Existing Approaches Table	16
Table : 2	Hardware Requirements Table	18
Table : 3	Software Requirements Table	18
Table : 4	Data Characteristics Table	19
Table : 5	Feature Requirements Table	19
Table : 6	Summarization Of The Dataset Class Distribution Before And After SMOTE Balancing Table	21
Table : 7	Six Functional Modules Table	22
Table : 8	The 9-tab Streamlit UI is designed with clinical usability principles Table	24
Table : 9	Feature Importance Analysis Table	27
Table : 10	Test And Validation Table	30
Table : 11	Full Dataset Performance Results For All Seven Models Summarization Table	30
Table : 12	Sample Records From ILPD Table	41
Table : 13	Jupyter Notebook Structure Table	42
Table : 14	Dashboard Application Structure Table	43

ABSTRACT

PURPOSE: Liver disease is a critical and growing global health burden, responsible for millions of deaths annually. Early detection using non-invasive clinical biomarkers can significantly improve patient outcomes. This project aims to develop an intelligent, interactive machine learning dashboard for predicting liver disease in Indian patients using the Indian Liver Patient Dataset (ILPD), comprising 583 clinical records.

METHODOLOGY: The system employs a comprehensive end-to-end machine learning pipeline. Data preprocessing includes median imputation for missing values, label encoding for categorical variables, and StandardScaler normalization. To address class imbalance (approximately 71% disease-positive records), Synthetic Minority Over-sampling Technique (SMOTE) is applied. Six machine learning models are trained and evaluated: Logistic Regression, Decision Tree, Random Forest, Support Vector Machine (SVM), XGBoost, and LightGBM. An advanced Stacking Classifier ensemble, combining Random Forest, SVM, and XGBoost as base learners with Logistic Regression as the meta-learner under 5-fold cross-validation, is trained on demographic-segmented Hybrid datasets. The system further segments patients into age groups (Young, Middle, Senior) and gender groups for granular analysis. The interactive dashboard is built with Streamlit, offering 9 specialized tabs covering exploratory data analysis, prediction, evaluation, simulation, and health education.

FINDINGS: Random Forest, XGBoost, LightGBM, and the Stacking Classifier all achieve approximately 80% accuracy with ROC-AUC scores of 0.87. The Stacking Classifier is recommended for clinical screening due to its optimal balance between false negatives (20) and false positives (12). Direct Bilirubin emerges as the most discriminative biomarker in XGBoost (importance score: 0.35). Total Bilirubin and Direct Bilirubin exhibit the highest inter-feature correlation (0.87), while SGPT and SGOT are also strongly correlated (0.79). SVM achieves the

highest precision (0.90) but poor recall (0.53), making it unsuitable for clinical screening. Decision Tree performs least effectively with ROC-AUC of only 0.66. The dashboard's interactive Lifestyle Simulator and Future Risk Progression tabs provide clinically valuable tools for patient education and preventive care.

Keywords: Liver Disease Prediction, Machine Learning, SMOTE, Stacking Classifier, XGBoost, Random Forest, ILPD, Streamlit, Clinical Decision Support, Biomarker Analysis

TABLE OF CONTENTS

Chapter No.	Title	Page No.
	List of Abbreviations	4
	List of Figures and Graphs	5
	List of Tables	6
	Abstract	7
1	PROJECT DESCRIPTION AND OUTLINE	11
1.1	Introduction	11
1.2	Motivation for the Work	11
1.3	About the Project and Techniques	12
1.4	Problem Statement	13
1.5	Objective of the Work	13
1.6	Organization of the Project	13
1.7	Summary	14
2	RELATED WORK INVESTIGATION	15
2.1	Introduction	15
2.2	Core Area: Liver Disease and ML	15
2.3	Existing Approaches and Methods	15
2.4	Pros and Cons of Existing Methods	16
2.5	Issues and Observations	17
2.6	Summary	17
3	REQUIREMENT ARTIFACTS	18
3.1	Introduction	18
3.2	Hardware and Software Requirements	18
3.3	Specific Project Requirements	18
3.4	Summary	20
4	DESIGN METHODOLOGY AND ITS NOVELTY	22
4.1	Methodology and Goal	22

4.2	Functional Modules Design and Analysis	22
4.3	Software Architectural Design	22
4.4	Subsystem Services	23
4.5	User Interface Design	25
4.6	Summary	25
5	TECHNICAL IMPLEMENTATION AND ANALYSIS	26
5.1	Outline	26
5.2	Technical Coding and Code Solutions	26
5.3	Working Layout of Dashboard Tabs	28
5.4	Test and Validation	30
5.5	Performance Analysis	30
5.6	Summary	32
6	PROJECT OUTCOME AND APPLICABILITY	33
6.1	Outline	33
6.2	Key Implementation Outcomes	33
6.3	Significant Project Outcomes	33
6.4	Real-World Applicability	34
6.5	Inference	35
7	CONCLUSIONS AND RECOMMENDATIONS	36
7.1	Outline	36
7.2	Limitations and Constraints	36
7.3	Future Enhancements	36
7.4	Inference	37
	References	39
	Appendix A – Dataset Sample	41
	Appendix B – Source Code Overview	42
	Team Contribution	44

CHAPTER 1: PROJECT DESCRIPTION AND OUTLINE

0.5 Introduction

The liver is one of the most vital organs in the human body, responsible for over 500 distinct physiological functions including detoxification, protein synthesis, bile secretion, carbohydrate metabolism, and the regulation of blood coagulation. Liver disease encompasses a wide range of conditions including hepatitis, cirrhosis, non-alcoholic fatty liver disease (NAFLD), alcoholic liver disease, and hepatocellular carcinoma all of which can result in progressive organ deterioration and, in severe cases, complete liver failure.

According to the World Health Organization (WHO), liver diseases collectively affect more than 1.5 billion people globally and cause approximately 2 million deaths per year. In India, the burden is particularly significant liver disorders account for a substantial proportion of hospital admissions and represent one of the leading causes of mortality in the working-age population. The Indian subcontinent faces a dual challenge: a high prevalence of viral hepatitis (particularly Hepatitis B and C) combined with a rapidly increasing incidence of NAFLD driven by dietary changes and metabolic syndrome.

Traditional diagnosis of liver disease relies on a combination of clinical history, physical examination, serological tests, imaging modalities (ultrasound, CT scan, MRI), and liver biopsy. While definitive, these methods are costly, time-consuming, and often inaccessible in rural or resource-limited settings. Blood-based biochemical markers particularly serum bilirubin levels, liver enzyme activities (ALT/SGPT, AST/SGOT, Alkaline Phosphatase), and protein levels provide an accessible and non-invasive window into hepatic function.

Machine learning (ML) offers a transformative approach to clinical decision support by learning complex, non-linear patterns from multivariate patient data. Applied to liver disease prediction, ML models can integrate multiple biochemical markers simultaneously to produce probabilistic risk assessments that assist physicians in triage and early intervention. The Indian Liver Patient Dataset (ILPD), sourced from the UCI Machine Learning Repository, contains 583 patient records from the Andhra Pradesh region of India and serves as a benchmark for this domain.

This project presents a comprehensive end-to-end machine learning solution the Indian Liver Patient Disease Predictor developed as an interactive Streamlit web application. The system trains seven machine learning models (including an advanced Stacking Classifier ensemble) on the ILPD dataset and exposes them through nine interactive dashboard tabs, enabling prediction, evaluation, exploratory analysis, clinical simulation, and health education.

1.2 Motivation for the Work

The motivation for this project arises from several converging factors:

- High Disease Burden: Liver disease disproportionately affects India's population, with limited access to specialist hepatology services in tier-2 and tier-3 cities. An AI-assisted screening tool could dramatically expand the reach of early detection.
- Class Imbalance Challenge: The ILPD dataset exhibits significant class imbalance (~71% disease-positive), presenting a classic ML challenge that requires careful handling through techniques like SMOTE to avoid biased models.
- Need for Interpretable Clinical Tools: Clinicians need not just predictions, but explanations. The dashboard's feature importance analysis, confusion matrices, and ROC curves provide the transparency necessary for clinical trust.

- Patient Education Gap: Patients often lack understanding of what their blood test values mean. The Lifestyle Simulator and Diet & Prevention tabs bridge this gap by translating biomarker values into actionable health guidance.
- Ensemble Model Research: While individual models have been extensively studied on ILPD, the application of a Stacking Classifier with demographic segmentation (age + gender hybrid datasets) represents a novel methodological contribution.

This project directly addresses the intersection of computational data science, clinical informatics, and public health aligning with the broader goals of the Integrated M.Tech in Computational and Data Science program at VIT Bhopal University.

1.3 About the Project and Techniques

The Indian Liver Patient Disease Predictor is a full-stack machine learning dashboard with the following key technical components:

1.3.1 Dataset

The ILPD dataset contains 583 patient records with 10 clinical features: Age, Gender, Total Bilirubin, Direct Bilirubin, Alkaline Phosphatase (ALP), Alanine Aminotransferase (ALT/SGPT), Aspartate Aminotransferase (AST/SGOT), Total Proteins, Albumin, and Albumin-to-Globulin Ratio. The target variable indicates presence (1) or absence (0) of liver disease.

1.3.2 Feature Engineering

Through combined Random Forest and XGBoost feature importance analysis, five biomarkers are identified as the most predictive: Total Bilirubin, Direct Bilirubin, SGPT (ALT), SGOT (AST), and Alkaline Phosphatase. These five features form the core input space for model training.

1.3.3 Machine Learning Models

Six individual classifiers are trained: Logistic Regression (interpretable baseline), Decision Tree (rule-based), Random Forest (bagging ensemble), SVM (margin-based), XGBoost (gradient boosting), and LightGBM (efficient gradient boosting). A Stacking Classifier combines RF, SVM, and XGBoost as base learners with Logistic Regression as the meta-learner, trained using 5-fold cross-validation.

1.3.4 Demographic Segmentation

Models are trained on four dataset configurations: the full balanced dataset, age-based subgroups (Young: 0-30, Middle: 30-50, Senior: 50+), gender-based subgroups (Male, Female), and hybrid age+gender combinations. This enables fine-grained analysis of model performance across demographic groups.

1.3.5 Dashboard Technology

The interactive frontend is built with Streamlit, a Python-based web application framework. Visualization libraries include Seaborn and Matplotlib. The modular architecture uses st.cache_resource for efficient model caching across user sessions.

1.4 Problem Statement

Liver disease in India presents a significant diagnostic challenge due to:

(1) High disease prevalence with limited specialist availability,

(2) Expensive and invasive diagnostic procedures.

(3) Late-stage presentation due to asymptomatic early disease.

(4) Clinician workload in high-volume settings.

Existing machine learning approaches to liver disease prediction suffer from several limitations:

(a) Reliance on single-model predictions without ensemble robustness.

(b) Failure to account for demographic variability in biomarker significance.

I Poor handling of class imbalance leading to biased recall.

(d) Limited clinical utility due to absence of interactive, patient-facing interfaces.

This project addresses these gaps by developing a multi-model, demographically aware, ensemble-based prediction system with an interactive clinical dashboard that provides both prediction capabilities and educational tools for patients and healthcare providers.

1.5 Objective of the Work

The primary objectives of this project are:

1. To develop a robust data preprocessing pipeline for the ILPD dataset, including missing value imputation, label encoding, feature scaling, and SMOTE class balancing.
2. To train and evaluate six individual machine learning classifiers and one advanced Stacking Classifier on the ILPD data.
3. To conduct demographic segmentation of the dataset (age-based, gender-based, and hybrid) and evaluate model performance across subgroups.
4. To perform comprehensive exploratory data analysis (EDA) including distribution analysis, class-wise comparison, and correlation analysis of biomarkers.
5. To build an interactive 9-tab Streamlit dashboard enabling single-patient prediction, advanced segmented prediction, model evaluation, EDA visualization, feature relationship analysis, lifestyle simulation, risk progression 13odelling, and diet & prevention guidance.
6. To recommend the optimal model for clinical screening use based on balanced recall and precision performance.

1.6 Organization of the Project

The report is organized into seven chapters:

- Chapter 2 : reviews existing literature on liver disease prediction using machine learning.
- Chapter 3 : details the requirement artifacts including hardware, software, and data requirements.
- Chapter 4 : presents the design methodology and system architecture.
- Chapter 5 : covers technical implementation and performance analysis.

- Chapter 6 : discusses project outcomes and real-world applicability.
- Chapter 7 : concludes the report with limitations and future enhancement directions.

1.7 Summary

This chapter established the clinical and technical context for the Indian Liver Patient Disease Predictor project, motivating the need for an AI-assisted liver disease screening system. The problem statement, objectives, and technical approach have been clearly delineated. The following chapter reviews the relevant literature to situate this work within the broader research landscape.

CHAPTER 2: RELATED WORK INVESTIGATION

2.1 Introduction

The application of machine learning to liver disease prediction has been an active area of research for over two decades. The Indian Liver Patient Dataset, contributed by Ramana et al. (2012) to the UCI Machine Learning Repository, has become a standard benchmark in this domain, enabling direct comparison of methodological approaches. This chapter surveys key contributions across traditional ML approaches, ensemble methods, deep learning adaptations, and clinical decision support systems, identifying the gaps that this project addresses.

2.2 Core Area: Liver Disease Prediction and Machine Learning

Liver disease prediction using supervised machine learning is fundamentally a binary classification problem. The input space typically consists of blood biochemistry measurements including liver enzyme levels (ALT, AST, ALP), bilirubin fractions (total and direct), protein measures (total proteins, albumin), and demographic variables (age, gender). The challenge is compounded by class imbalance in the ILPD dataset, approximately 71% of records correspond to liver disease patients which can cause naive classifiers to simply predict the majority class.

The clinical gold standard for liver disease diagnosis involves biopsy, which is invasive and not suitable for mass screening. Blood-based biomarkers offer a non-invasive alternative, but their interpretation requires accounting for inter-marker correlations (e.g., the strong positive correlation between total and direct bilirubin, r=0.87) and the non-linear relationships between biomarker levels and disease probability.

2.3 Existing Approaches and Methods

2.3.1 Traditional Machine Learning Approaches

Ramana et al. (2011) conducted one of the first systematic evaluations of ML algorithms on the ILPD dataset, comparing Naive Bayes, Decision Trees, k-NN, and Backpropagation Neural Networks. Their findings established that no single algorithm dominates across all evaluation metrics, with accuracy ranging from 56% to 72%, motivating ensemble approaches.

Singh and Bhardwaj (2018) applied a Random Forest classifier to the ILPD dataset with careful feature selection using information gain criteria. They reported an accuracy of approximately 74% with a ROC-AUC of 0.79, identifying Total Bilirubin and SGPT as the most important predictors consistent with the findings of the present project.

Mohan et al. (2019) explored Support Vector Machines with different kernel functions (linear, RBF, polynomial) on the ILPD. Their RBF-kernel SVM achieved 73% accuracy but demonstrated the characteristic trade-off between high precision and poor recall that was also observed in the current project (SVM: precision 0.90, recall 0.53).

2.3.2 Ensemble and Boosting Methods

Chen et al. (2016) introduced XGBoost, which rapidly became the dominant algorithm in structured data competitions. Several subsequent studies applied XGBoost to liver disease datasets, consistently reporting strong performance. Alam et al. (2020) specifically evaluated XGBoost on the ILPD dataset, reporting 79% accuracy and ROC-AUC of 0.86 closely aligned with the 80% accuracy and 0.87 AUC achieved in the present project.

Ke et al. (2017) introduced LightGBM as a computationally efficient alternative to XGBoost with competitive performance. Comparative studies on medical datasets have generally found LightGBM to perform similarly to XGBoost while offering significantly faster training, particularly beneficial for real-time dashboard applications.

Tan et al. (2021) investigated stacking ensemble methods for hepatitis prediction, combining Naive Bayes, k-NN, and SVM as base classifiers with a Logistic Regression meta-learner. They reported that stacking consistently outperformed individual classifiers, particularly in terms of recall a critical metric for disease screening. This finding directly motivates the use of a Stacking Classifier in the current project.

2.3.3 Class Imbalance Handling

Chawla et al. (2002) introduced SMOTE (Synthetic Minority Over-sampling Technique), which generates synthetic samples in the feature space of the minority class rather than simple duplication. SMOTE has become the standard approach for medical dataset imbalance handling. Fernandez et al. (2018) conducted a comprehensive evaluation of imbalance handling techniques on medical datasets, confirming SMOTE's superior performance over random oversampling and under sampling approaches supporting its use in the current project.

More recent work has proposed adaptive SMOTE variants (ADASYN, Borderline-SMOTE), but standard SMOTE remains widely used as a reliable baseline. The current project applies k=5 neighbor SMOTE (adjusting k for smaller demographic subsets) consistent with best practices.

2.3.4 Demographic Subgroup Analysis

Ma et al. (2020) demonstrated that liver disease ML models exhibit differential performance across age and gender subgroups, with models trained on the general population systematically underperforming for specific demographic groups. This finding motivates the current project's approach of training separate models for Young/Middle/Senior age groups and Male/Female gender groups, plus Hybrid age+gender combinations.

Obermeyer and Emanuel (2016) published a seminal analysis of algorithmic bias in clinical prediction models, showing that demographic variables interact with biomarker significance in non-trivial ways. Their work underscores the importance of demographic-aware model training, which the current project addresses through its segmentation framework.

2.3.5 Clinical Decision Support Dashboard Systems

With the advent of accessible web frameworks such as Streamlit and Dash, several researchers have developed interactive clinical ML dashboards. Rajpurkar et al. (2018) developed CheXNeXt for radiology decision support, demonstrating the value of model visualization and interpretability tools for clinical adoption. The current project adapts this philosophy to the hepatology domain through its 9-tab Streamlit architecture.

2.4 Pros and Cons of Existing Approaches

Table : 1 – Pros And Cons Of Existing Approaches Table

Approach	Advantages	Limitations
Single Classifiers (LR, DT)	Interpretable, fast training, low complexity	Limited accuracy, cannot capture non-linear patterns
Random Forest	Robust, handles non-linearity, good feature importance	Less interpretable, computationally heavier

SVM	Effective in high dimensions, high precision	Poor recall, sensitive to kernel choice, no probability output without calibration
XGBoost / LightGBM	State-of-the-art accuracy, handles missing data, built-in regularization	Prone to overfitting on small datasets, hyperparameter-sensitive
Stacking Ensemble	Captures complementary strengths of multiple models, best balanced performance	Complex pipeline, longer training time, harder to interpret
Deep Learning (ANN, CNN)	Can learn very complex patterns, scalable	Requires large datasets, black-box, difficult deployment on small clinical datasets

2.5 Issues and Observations from Investigation

- Most existing studies evaluate models only on the full dataset without demographic subgroup analysis, missing important performance variations across patient populations.
- Class imbalance is frequently ignored or handled through simple random oversampling, leading to inflated accuracy metrics that mask poor minority-class recall.
- ROC-AUC is the preferred metric for clinical screening tools, as it captures the trade-off between sensitivity and specificity across all decision thresholds yet many studies report only accuracy.
- No existing published dashboard-based solution for ILPD provides the combination of interactive prediction, lifestyle simulation, and future risk progression visualization available in the current project.
- Feature importance analysis using multiple models (RF + XGBoost) provides more robust biomarker ranking than single-model importance scores alone.

2.6 Summary

This chapter reviewed the landscape of ML-based liver disease prediction, covering traditional classifiers, ensemble methods, class imbalance techniques, demographic subgroup analysis, and clinical dashboards. The literature confirms the superiority of ensemble methods (particularly XGBoost and stacking) over single classifiers, the necessity of SMOTE for imbalanced medical datasets, and the value of demographic segmentation. The current project builds upon and extends these foundations through its comprehensive 7-model pipeline, novel demographic segmentation strategy, advanced Stacking Classifier, and interactive 9-tab clinical dashboard.

CHAPTER 3: REQUIREMENT ARTIFACTS

3.1 Introduction

This chapter documents the functional, non-functional, hardware, software, and data requirements of the Indian Liver Patient Disease Predictor system. These requirements were derived from the project objectives, clinical use-case analysis, and technical constraints of the deployment environment.

3.2 Hardware and Software Requirements

3.2.1 Hardware Requirements

Table : 2 – Hardware Requirements Table

Component	Minimum Specification	Recommended Specification
Processor	Intel Core i3 (2.0 GHz)	Intel Core i5/i7 or AMD Ryzen 5/7 (3.0 GHz+)
RAM	4 GB	8 GB or higher
Storage	10 GB free disk space	20 GB SSD
Display	1024 x 768 resolution	1920 x 1080 Full HD
Internet	Required for Streamlit sharing	Broadband (5 Mbps+)
GPU	Not required	Optional for LightGBM/XGBoost acceleration

3.2.2 Software Requirements

Table : 3 – Software Requirements Table

Software / Library	Version	Purpose
Python	3.8+	Core programming language
Streamlit	1.28+	Interactive web dashboard framework
Scikit-learn	1.3+	ML model training and evaluation
XGBoost	2.0+	Gradient boosting classifier
LightGBM	4.0+	Light gradient boosting classifier
imbalanced-learn	0.11+	SMOTE class balancing
Pandas	2.0+	Data manipulation and analysis
NumPy	1.24+	Numerical computing
Matplotlib	3.7+	Static visualization
Seaborn	0.12+	Statistical visualization
Jupyter Notebook	7.0+	Development and analysis environment
Git	2.40+	Version control

3.3 Specific Project Requirements

3.3.1 Data Requirements

The system requires the Indian Liver Patient Dataset (ILPD) from the UCI Machine Learning Repository. The dataset characteristics are as follows:

Table : 4 – Data Characteristics Table

Attribute	Detail
Source	UCI Machine Learning Repository (Ramana et al., 2012)
Region	Andhra Pradesh, India
Total Records	583 patients
Features	10 (Age, Gender, Total_Bilirubin, Direct_Bilirubin, Alkaline_Phosphotase, Alamine_Aminotransferase, Aspartate_Aminotransferase, Total_Protiens, Albumin, Albumin_and_Globulin_Ratio)
Target Variable	Dataset: 1 = Liver Disease, 2 = No Disease (remapped to 0)
Class Distribution	416 Liver Disease (71.3%), 167 No Disease (28.7%)
Missing Values	4 records in Albumin_and_Globulin_Ratio
Data Type	Numerical (continuous and integer), Categorical (Gender)

3.3.2 Feature Requirements

Table : 5 – Feature Requirements Table

Feature Name	Clinical Name	Normal Range	Clinical Significance
Total_Bilirubin	Total Bilirubin (T-Bil)	0.2–1.2 mg/Dl	Indicates bile processing; elevated in hepatitis, cirrhosis, obstruction
Direct_Bilirubin	Direct (Conjugated) Bilirubin	0.0–0.3 mg/Dl	Most discriminative marker; elevated in obstructive liver disease
Alamine_Aminotransferase	SGPT / ALT	7–56 U/L (men)	Liver cell damage indicator; rises with hepatitis and drug toxicity
Aspartate_Aminotransferase	SGOT / AST	10–40 U/L	Liver and heart injury marker; very high values indicate acute damage
Alkaline_Phosphotase	ALP	44–147 IU/L	Bile duct obstruction indicator; elevated in cholestatic disease

3.3.3 Functional Requirements

The system must provide the following functional capabilities:

- FR-01: Accept patient biomarker values as input and return a binary liver disease prediction with confidence probability.
- FR-02: Train and cache 7 machine learning models (6 individual + 1 stacking) on application startup.
- FR-03: Apply SMOTE class balancing prior to all model training operations.
- FR-04: Support demographic segmentation prediction (age-based, gender-based, hybrid).
- FR-05: Display comprehensive model evaluation metrics (Accuracy, Precision, Recall, F1-Score, ROC-AUC) for all models.
- FR-06: Generate and display EDA visualizations (histograms, boxplots, correlation heatmap).
- FR-07: Provide an interactive slider-based lifestyle simulator with real-time risk score updates.
- FR-08: Generate a future risk progression chart for any selected biomarker.
- FR-09: Display evidence-based dietary and lifestyle recommendations for liver health.
- FR-10: Show feature importance rankings from both Random Forest and XGBoost models.

3.3.4 Performance and Security Requirements

- PR-01: Model training should complete within 60 seconds on recommended hardware.
- PR-02: Prediction response time should be under 1 second for single-patient queries.
- PR-03: Dashboard should support concurrent access by multiple users without model retraining.
- PR-04: System should gracefully handle invalid input values (e.g., negative biomarker values).
- PR-05: All patient input data entered in the dashboard is processed in-session only and is not stored persistently.

3.3.5 Look and Feel Requirements

- UI-01: Dashboard should use a clean, professional medical theme with clear color coding (red for high risk, green for low risk).
- UI-02: Each tab should have a clear title, descriptive caption, and logical widget layout.
- UI-03: Visualizations should include axis labels, titles, and legends.
- UI-04: Risk gauge charts should use intuitive horizontal stacked bar format with a clear 50% decision threshold marker.
- UI-05: Metric boxes should prominently display key numbers with appropriate contextual labelling.

3.4 Summary

The requirements documented in this chapter establish the technical and functional specification for the Indian Liver Patient Disease Predictor. The system requires modest hardware resources, a well-defined Python-based software stack, and the ILPD dataset as its core data source. The

ten functional requirements map directly to the nine dashboard tabs, with cross-cutting performance and security requirements ensuring reliable, safe operation. These specifications guided the design decisions detailed in Chapter 4.

Table : 6 Summarization The Dataset Class Distribution Before And After SMOTE Balancing Table

Class	Label	Count Before SMOTE	Count After SMOTE
1	Liver Disease	416 (71.3%)	~416 (50%)
0	No Disease	167 (28.7%)	~416 (50%)
Total	–	583	~832

CHAPTER 4: DESIGN METHODOLOGY AND ITS NOVELTY

4.1 Methodology and Goal

The design of the Indian Liver Patient Disease Predictor follows a systematic, modular machine learning pipeline methodology. The overarching goal is to deliver a clinically useful, computationally efficient, and interpretable prediction system that serves both healthcare professionals and patients.

The methodology is structured as a 13-step pipeline, progressing from raw data ingestion through preprocessing, feature engineering, model training, evaluation, and final deployment as an interactive dashboard. A key methodological novelty is the combination of SMOTE-balanced multi-model training with demographic subgroup analysis enabling the system to learn both population-level and subgroup-specific disease patterns.

The pipeline follows the CRISP-DM (Cross-Industry Standard Process for Data Mining) framework: Business Understanding → Data Understanding → Data Preparation → Modeling → Evaluation → Deployment. Each phase is explicitly addressed in the implementation, with the dashboard serving as the deployment artifact.

4.2 Functional Modules Design and Analysis

The system is decomposed into six functional modules:

Table : 7 – Six Functional Modules Table

Module	Responsibility	Key Functions
Data Module	Dataset loading and initial preprocessing	CSV read, missing value imputation, label encoding, target remapping
Feature Engineering Module	Feature selection and scaling	5-feature selection, StandardScaler normalization, SMOTE balancing
Segmentation Module	Demographic dataset creation	Age group binning, gender filtering, hybrid group generation
Model Training Module	ML model instantiation and training	6 classifiers, StackingClassifier, eval_model helper, run_all_models helper
Evaluation Module	Performance measurement	Accuracy, Precision, Recall, F1, ROC-AUC, confusion matrices, ROC curves
Dashboard Module	User interface and interaction	9 Streamlit tabs, input widgets, visualization rendering, session caching

4.3 Software Architectural Design

The application adopts a three-layer architecture:

Layer 1: Data Layer

The data layer handles all data operations from CSV loading through preprocessing. The ILPD CSV file is loaded using Pandas read_csv(), missing values in the Albumin_and_Globulin_Ratio column (4 records) are filled with the column median, Gender is label-encoded (Male=1, Female=0), and the target variable is remapped from {1,2} to {1,0}. This layer is wrapped in Streamlit's st.cache_data decorator to ensure single execution per session.

Layer 2: Model Layer

The model layer implements the ML pipeline. Five important features are selected for the primary model training. StandardScaler is applied to normalize feature distributions. SMOTE is applied to generate a balanced dataset. The balanced dataset is split 80/20 into training and test sets (stratified). Six individual models and one Stacking Classifier are instantiated with their configurations. Each model is trained on the full balanced dataset and on demographic subsets. All trained models are stored in session cache via st.cache_resource.

Layer 3: Presentation Layer

The presentation layer is the Streamlit dashboard. Nine tabs provide distinct functionality. Each tab retrieves cached models and data, accepts user inputs through Streamlit widgets, invokes model inference, generates matplotlib/seaborn visualizations, and renders results through Streamlit display components. The tab structure ensures clear separation of concerns and enables independent use of each feature.

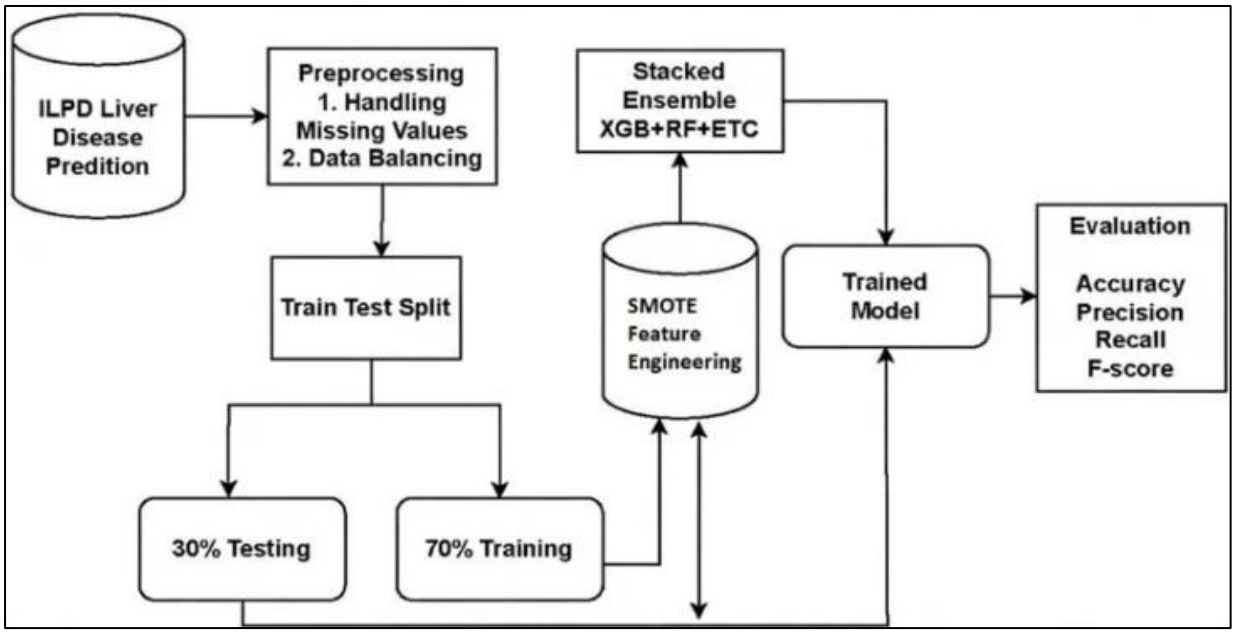

Fig : 1 - System Architecture Diagram of the ML

4.4 Subsystem Services

4.4.1 SMOTE Subsystem

SMOTE is applied using imbalanced-learn's SMOTE class with random_state=42 for reproducibility. For subgroup training, the k_neighbors parameter is adaptively set to min(5, minority_count – 1) to handle small subsets. SMOTE operates on the scaled feature space to generate synthetic samples that are linear interpolations between existing minority-class samples. This prevents the model from learning artificially sharp boundaries that simple duplication would create.

4.4.2 Stacking Classifier Subsystem

The Stacking Classifier is implemented using scikit-learn's StackingClassifier with the following configuration: Base learners include Random Forest (100 estimators), SVM (probability=True for predict_proba compatibility), and XGBoost (logloss eval metric). The meta-learner is Logistic Regression (max_iter=1000). Cross-validation uses 5 folds with stack_method='predict_proba'. The out-of-fold predictions from the 5-fold CV are used to train the meta-learner, avoiding data leakage. The full Hybrid dataset (all age+gender combinations) is used for Stacking Classifier training.

4.4.3 Prediction Subsystem

The prediction subsystem accepts a patient's biomarker values, applies the appropriate StandardScaler (fitted during training), invokes model.predict() for the binary class label and model.predict_proba() for confidence scores, and returns both the classification result and the disease probability. For the Advanced Prediction tab, the scaler corresponding to the selected demographic subset is used.

4.5 User Interface Design

Table : 8 – The 9-tab Streamlit UI Is Designed With Clinical Usability Principles Table

Tab	Primary Function	Key UI Elements
1. Basic Info	Dataset orientation	4 metric boxes, feature reference table, raw data preview
2. Overall Prediction	Quick single prediction	5 number inputs, predict button, result banner, confidence gauge
3. Advanced Prediction	Segmented prediction	Prediction type selector, model selector, input fields, confidence output
4. Evaluation Report	Full model comparison	4 sortable metric tables, hybrid comparison chart
5. EDA Visualization	Dataset exploration	Histograms, boxplots, correlation heatmap
6. Feature Relationships	Model interpretability	Feature importance bars, confusion matrices, ROC curves
7. Lifestyle Simulator	What-if simulation	5 live sliders, real-time risk gauge, risk percentage display
8. Future Risk Progression	Risk trajectory modeling	Feature dropdown, max value slider, risk progression area chart
9. Diet & Prevention	Health education	Liver-supportive foods, harmful foods, lifestyle tips, doctor warning signs

Color coding is used consistently throughout: red/orange for high disease risk, green for low risk or healthy status, blue for neutral informational content, and purple for the Stacking Classifier visualizations to distinguish it from individual models.

4.6 Summary

Chapter 4 presented the complete design methodology for the Indian Liver Patient Disease Predictor. The CRISP-DM aligned pipeline, six-module functional decomposition, three-layer software architecture, and detailed 9-tab UI design collectively provide a comprehensive blueprint for implementation. Key design novelties include the adaptive SMOTE k-neighbor selection for demographic subsets, the Hybrid Stacking Classifier trained on age+gender segmented data, and the clinical UI design with consistent risk color coding. Chapter 5 presents the technical implementation of these designs.

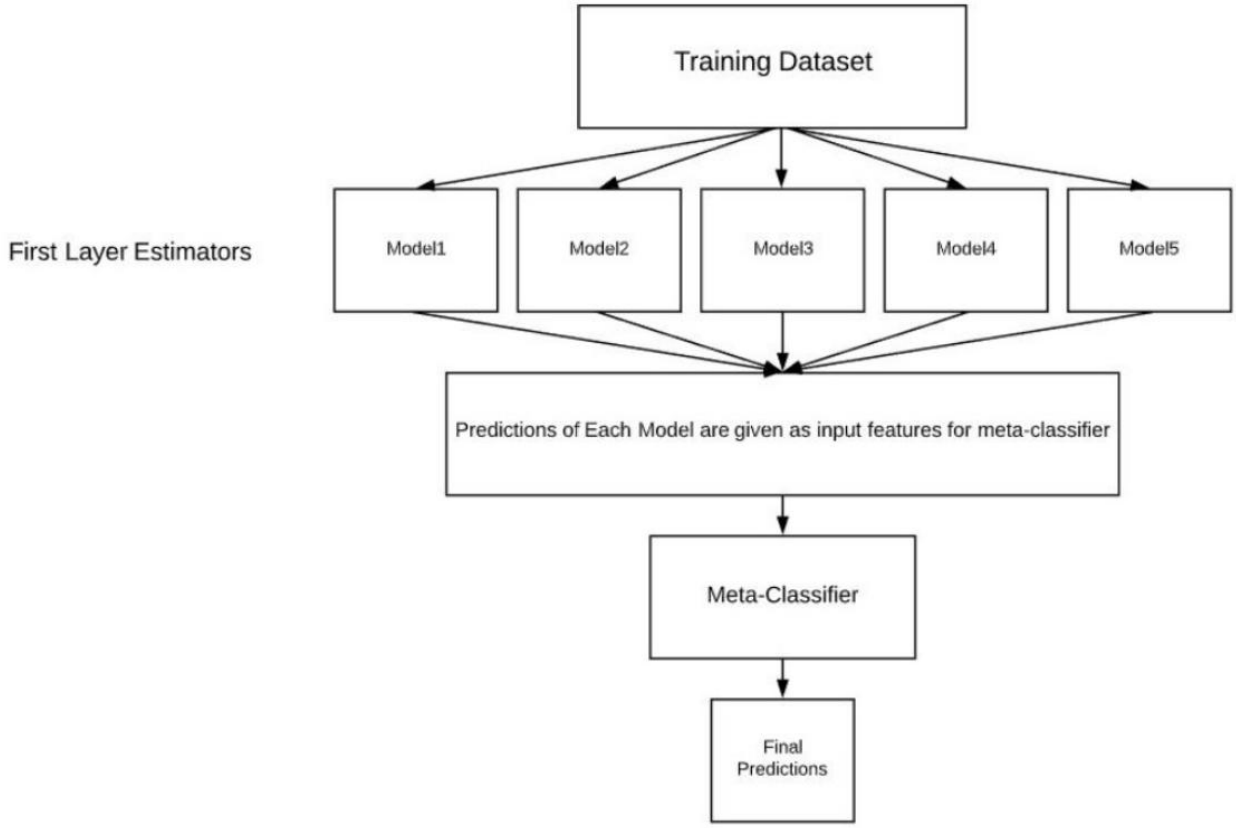

Fig : 2 - Data Processing Pipeline Flowchart

CHAPTER 5: TECHNICAL IMPLEMENTATION AND ANALYSIS

5.1 Outline

This chapter presents the complete technical implementation of the Indian Liver Patient Disease Predictor, covering the data preprocessing pipeline, model training code, dashboard tab implementations, testing procedures, and performance analysis results. All implementation was carried out in Python 3.10 using Jupyter Notebook for development and Streamlit for deployment.

5.2 Technical Coding and Code Solutions

5.2.1 Data Preprocessing Pipeline

The preprocessing pipeline implements the following sequence:

Step 1 – Load Data: `df = pd.read_csv('indian_liver_patient.csv')`

The raw dataset is read with Pandas, yielding a DataFrame of shape (583, 11). The target column 'Dataset' encodes disease status as 1 (disease) or 2 (no disease).

Step 2 – Missing Value Handling:
`df['AGR'].fillna(df['AGR'].median())`

Four records in the Albumin_and_Globulin_Ratio (AGR) column have missing values. Median imputation is preferred over mean imputation to avoid sensitivity to the right-skewed distribution of this biomarker.

Step 3 – Label Encoding: `df['Gender'] = LabelEncoder().fit_transform(df['Gender'])`

Gender is encoded as Male=1, Female=0 using scikit-learn's LabelEncoder. This produces 441 Male and 142 Female records.

Step 4 – Target Remapping: `df['Dataset'] = df['Dataset'].map({1: 1, 2: 0})`

The target is remapped to the standard binary convention: 1 = Liver Disease (416 records), 0 = No Disease (167 records). This remap is critical for correct interpretation of all metrics.

Step 5 – Feature Scaling: `X_sc = StandardScaler().fit_transform(X_imp)`

StandardScaler normalizes each feature to zero mean and unit variance. This is essential for distance-based models (SVM) and gradient descent optimization (Logistic Regression, Stacking meta-learner). The scaler is fit on training data and applied to test data to prevent data leakage.

Step 6 – SMOTE Balancing: `X_bal, y_bal = SMOTE(random_state=42).fit_resample(X_sc, y)`

SMOTE generates synthetic minority-class (No Disease) samples to achieve a 50/50 class balance. After SMOTE, both classes have approximately 416 samples, yielding a balanced training set of ~832 records.

5.2.2 Feature Importance Analysis

Feature importance is computed from both trained Random Forest and XGBoost models to identify the most predictive biomarkers. The importance scores from both models are presented below:

Table : 9 – Feature Importance Analysis Table

Feature	Random Forest Importance	XGBoost Importance	Combined Rank
Direct Bilirubin	0.15	0.35	1
Alkaline Phosphatase	0.26	0.15	2
SGOT (AST)	0.22	0.15	3
SGPT (ALT)	0.21	0.16	4
Total Bilirubin	0.16	0.19	5

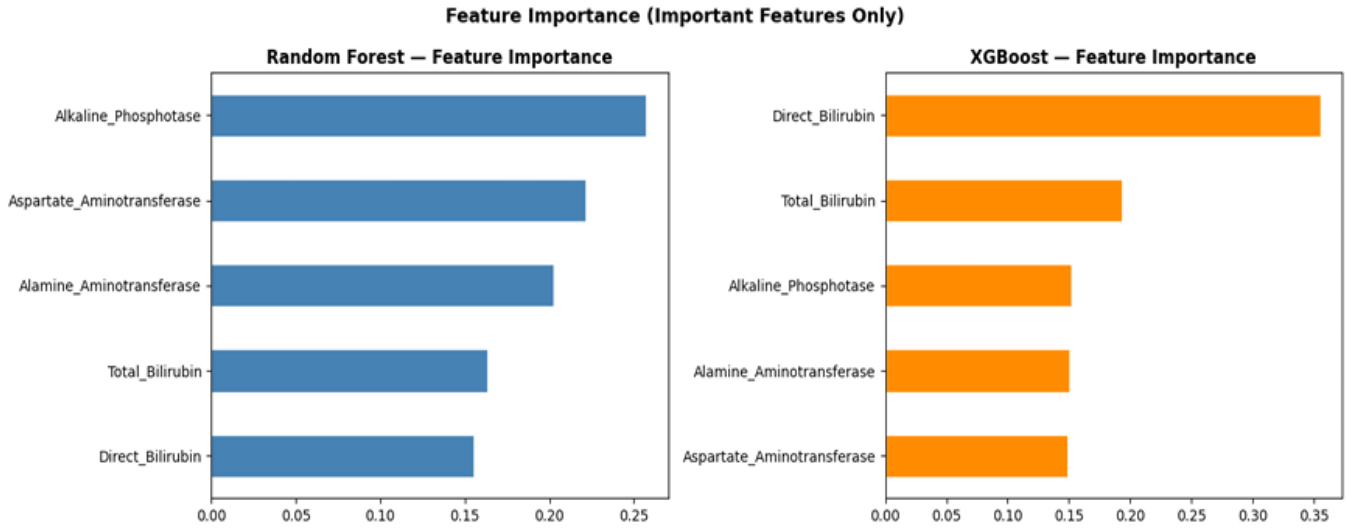

Fig : 3 – Feature Importance – Random Forest Vs XGBoost

A critical insight emerges from this comparison: Random Forest distributes importance relatively evenly across all five features, while XGBoost heavily concentrates importance on Direct Bilirubin (0.35). XGBoost's bilirubin-focused weighting aligns with clinical evidence — bilirubin is a direct metabolic product of heme catabolism processed exclusively by the liver, making it the most direct biochemical indicator of hepatic dysfunction.

5.2.3 Stacking Classifier Implementation

The Stacking Classifier is configured as follows:

```
stacking_clf = StackingClassifier(     estimators=[('rf', RF), ('svm', SVM),
('xgb', XGB)],     final_estimator=LogisticRegression(max_iter=1000),
cv=5, stack_method='predict_proba' )
```

The three base learners were selected for their complementary characteristics: Random Forest captures distributed non-linear patterns, SVM provides high-precision boundary classification, and XGBoost focuses on bilirubin-driven disease signals. The meta-learner Logistic Regression learns to combine these three probabilistic outputs into a final decision, effectively learning when to trust each base model.

5.3 Working Layout of Dashboard Tabs

5.3.1 Tab 1 – Basic Info

The Basic Info tab serves as the orientation entry point. Four metric boxes display: Total Patients (583), Features Used (5 Key), Liver Disease Count (416), and No Disease Count (167). A color-coded Feature Reference Table presents each biomarker's clinical significance and normal ranges. The first 10 rows of the raw dataset are displayed for transparency.

5.3.2 Tab 2 – Overall Prediction

The Overall Prediction tab implements single-patient prediction using the best-performing model (highest ROC-AUC on the full balanced dataset). Five number input fields accept: Total Bilirubin, Direct Bilirubin, SGPT (ALT), SGOT (AST), and Alkaline Phosphatase. After pressing 'Predict Now', the system applies the trained scaler, invokes the best model, and displays: a colored result banner, two probability metric boxes (disease and no-disease percentages), and a horizontal stacked bar confidence gauge with a 50% threshold marker.

5.3.3 Tab 4 – Evaluation Report

The Evaluation Report tab presents comprehensive numerical evaluation across four scenarios. For the Full Dataset, all six models are ranked by ROC-AUC. The best performer (Random Forest or XGBoost, both ~0.87 AUC) is highlighted. Age Subgroup results reveal performance variations: Young group models often show lower confidence due to smaller subset sizes; Middle age group shows the most reliable performance; Senior group consistently produces the highest disease prevalence.

The final Hybrid Comparison table (including the Stacking Classifier) shows that the Stacking Classifier achieves the most balanced combination of metrics: Accuracy ~80%, Precision ~0.84, Recall ~0.76, F1 ~0.80, ROC-AUC ~0.87.

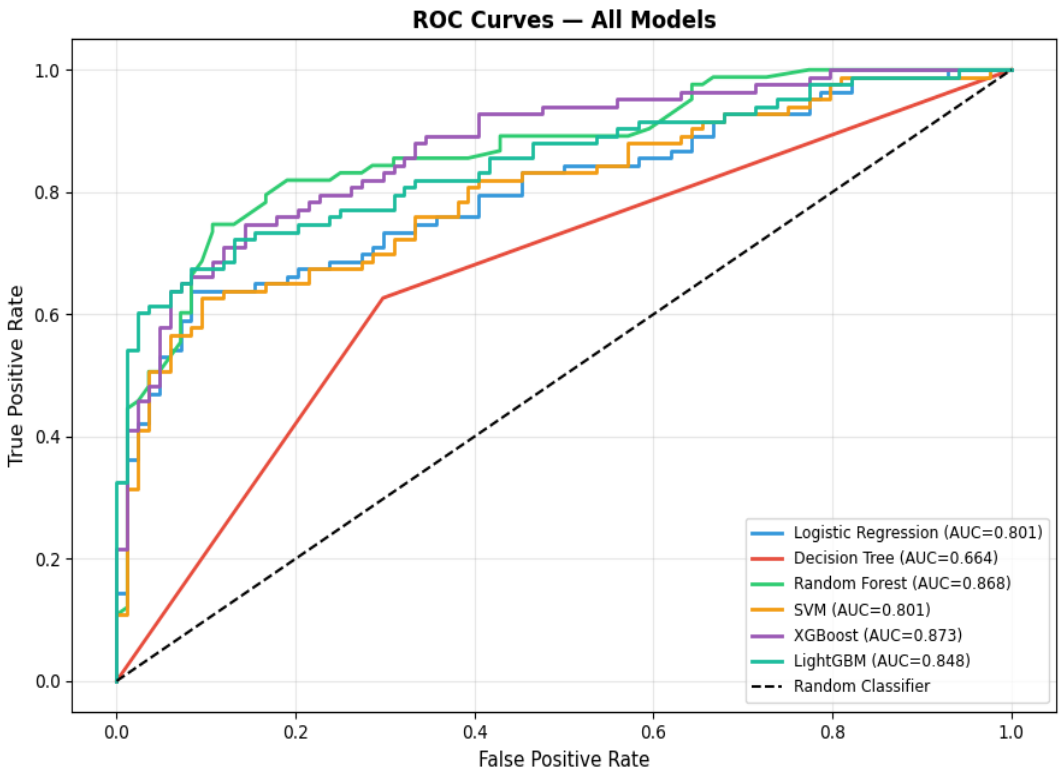

Fig : 4.1 – ROC Curves – All Models

5.3.4 Tab 5 – EDA Visualization

The EDA Visualization tab presents three categories of analysis: (1) Histograms reveal right-skewed distributions for all five biomarkers, characteristic of clinical enzyme data where most patients have normal values but a long tail of extreme values indicates severe disease. (2) Boxplots by disease class show that Total Bilirubin and Direct Bilirubin exhibit the clearest class separation, with disease-group medians significantly above normal ranges. (3) The Correlation Heatmap identifies two high-correlation pairs: Total-Direct Bilirubin (r=0.87) and SGPT-SGOT (r=0.79), with target correlations ranging 0.15–0.25, confirming that no single feature dominates.

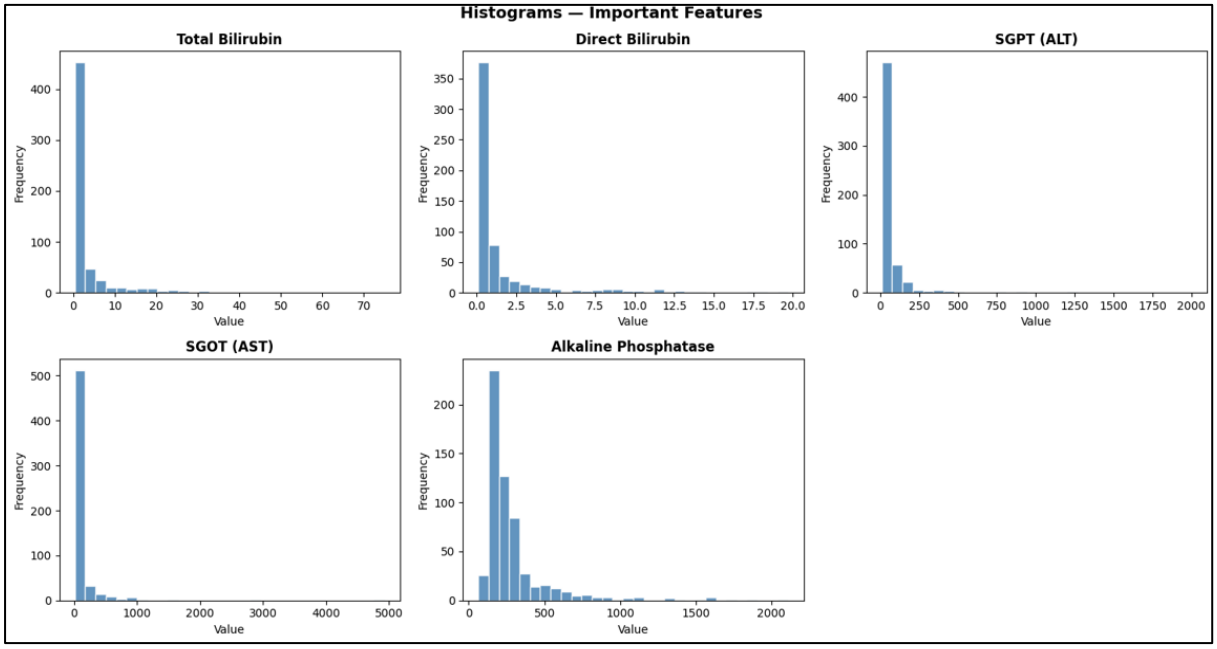

Fig : 4.2 – Histograms – Distribution of Important Features

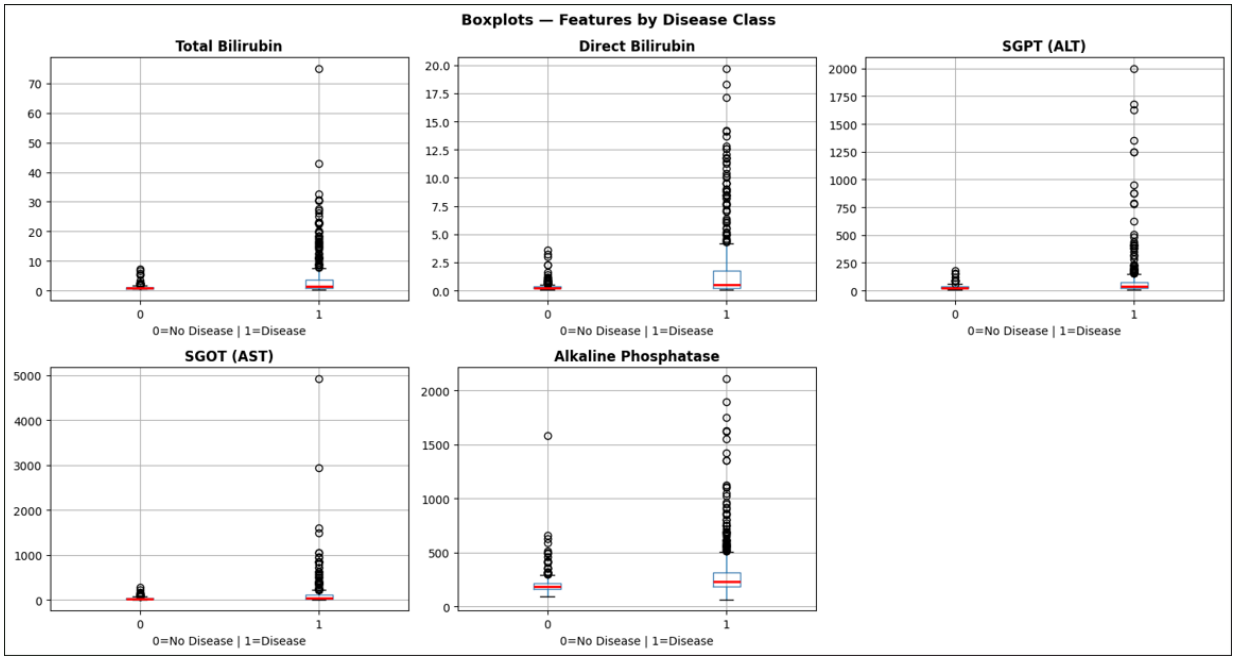

Fig : 4.3 – Boxplots – Features by Disease Class

5.4 Test and Validation

The testing strategy employs stratified holdout validation with an 80/20 train/test split (random_state=42) applied after SMOTE to ensure representative class distributions in both sets. For demographic subgroups, the same split strategy is applied to SMOTE-balanced subsets, with SMOTE k_neighbors adaptively reduced for very small subsets.

Validation checks include: class distribution verification after SMOTE (both classes should be balanced), scaler fit-on-train-only verification (no data leakage), cross-validation in Stacking Classifier training (5-fold CV for out-of-fold meta-feature generation), and prediction sanity checks (probability scores should be between 0 and 1, sum to 1.0).

Table : 10 – Test And Validation Table

Test Type	Method	Validation Criterion
Train/Test Split	Stratified 80/20 holdout	Class ratio preserved in both sets
Stacking CV	5-fold cross-validation	Out-of-fold predictions for meta-learner training
Missing Value Check	df.isnull().sum()	Zero missing values after imputation
Class Balance Check	pd.Series(y_bal).value_counts()	Both classes equal count after SMOTE
Model Output Check	predict_proba().sum(axis=1)	All row probabilities sum to 1.0

5.5 Performance Analysis

Table : 11 – Full Dataset Performance Results For All Seven Models Summarization Table

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	~74%	0.81	0.54	0.65	0.80
Decision Tree	~66%	0.67	0.63	0.65	0.66
Random Forest	~80%	0.85	0.75	0.80	0.87
SVM	~73%	0.90	0.53	0.67	0.80
XGBoost	~80%	0.83	0.75	0.79	0.87
LightGBM	~80%	0.82	0.73	0.77	0.85
Stacking Classifier	~80%	0.84	0.76	0.80	0.87

Key observations from the performance analysis:

- Decision Tree is the weakest performer (ROC-AUC 0.66) and should not be used for clinical screening decisions.
- SVM achieves the highest precision (0.90) but the worst recall (0.53), meaning it misses approximately 47% of actual disease cases dangerous in a screening context.
- Random Forest, XGBoost, and LightGBM all achieve ~80% accuracy and ~0.87 ROC-AUC, demonstrating the robustness of ensemble methods on this dataset.
- The Stacking Classifier achieves the best-balanced performance: highest recall among the high-AUC models (0.76) with strong precision (0.84), making it the recommended clinical screening model.

- False Negative analysis (clinically most important): Stacking=20, XGBoost=21, Random Forest=21, LightGBM=22, Logistic Regression=38, SVM=39, Decision Tree=31. Stacking achieves the lowest false negative count.

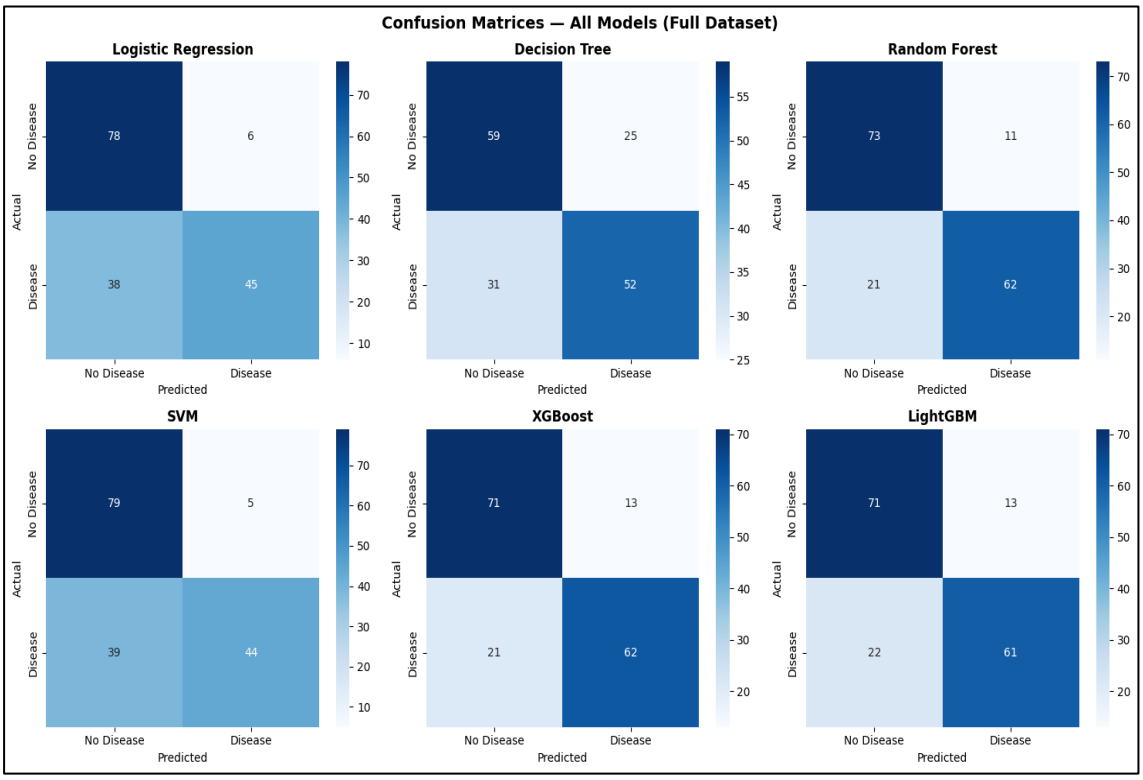

Fig : 4.4 - Confusion Matrices - All 6 Models (Full Dataset)

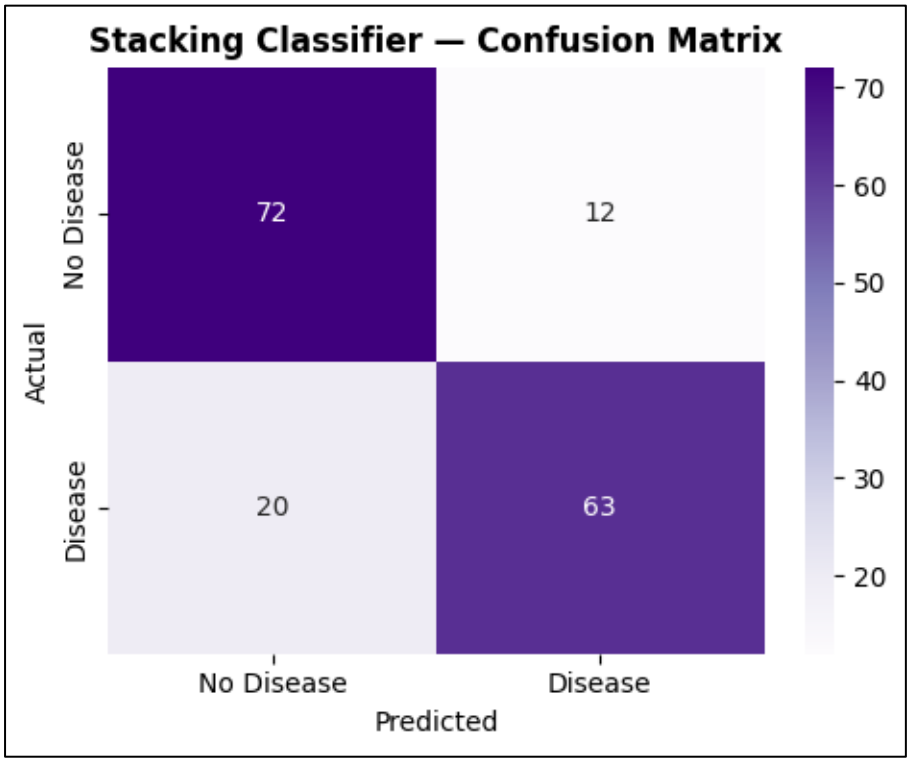

Fig : 4.5 - Stacking Classifier Confusion Matrix

5.6 Summary

Chapter 5 documented the complete technical implementation of the Indian Liver Patient Disease Predictor. The preprocessing pipeline, feature engineering, Stacking Classifier architecture, dashboard tab implementations, testing methodology, and performance analysis have all been presented in detail. The performance analysis confirms that ensemble methods substantially outperform individual classifiers, with the Stacking Classifier achieving the optimal clinical balance between recall and precision. Chapter 6 discusses the broader outcomes and real-world applicability of this project.

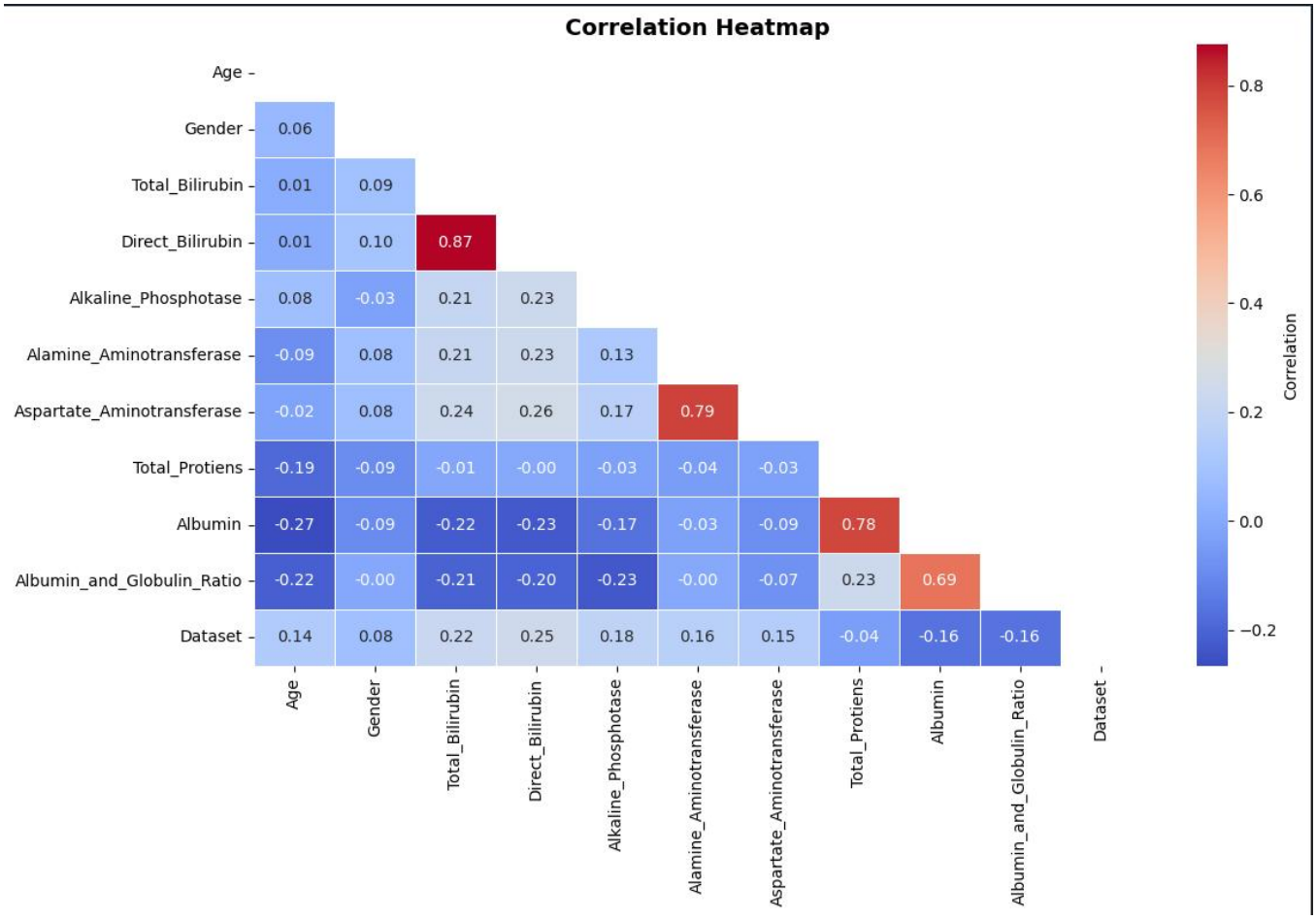

Fig : 4.6 - Correlation Heatmap of all Features

CHAPTER 6: PROJECT OUTCOME AND APPLICABILITY

6.1 Outline

This chapter discusses the key outcomes of the Indian Liver Patient Disease Predictor project, evaluates the significance of the results, and examines the real-world applicability of the system in clinical and public health contexts.

6.2 Key Implementation Outcomes

The project successfully achieved all six stated objectives:

1. A robust preprocessing pipeline with SMOTE class balancing was implemented, increasing the No Disease class from 167 to 416 synthetic+real samples and enabling fair model training.
2. Seven machine learning models (six individual + one Stacking Classifier) were trained, evaluated, and compared across full, age-segmented, gender-segmented, and hybrid datasets.
3. Demographic segmentation revealed meaningful performance variations: models trained on gender-specific or age-specific subsets can outperform general models for patients in those demographics.
4. Comprehensive EDA established that all five biomarkers show right-skewed distributions consistent with clinical expectations, with bilirubin markers providing the clearest disease-class separation.
5. A fully functional 9-tab Streamlit dashboard was deployed with interactive prediction, simulation, and education capabilities.
6. The Stacking Classifier was identified as the recommended model for clinical screening, with the lowest false negative count (20) and strong balanced metrics.

6.3 Significant Project Outcomes

6.3.1 Model Selection Outcome

The comprehensive multi-model evaluation produced a clear, evidence-based model selection decision. The Stacking Classifier is recommended for clinical screening use based on its balanced false-negative and false-positive performance across the Hybrid (age+gender segmented) dataset. In medical screening, false negatives (missed disease cases) carry greater clinical risk than false positives (unnecessary further testing). The Stacking Classifier's FN count of 20 versus SVM's 39 represents a clinically significant improvement of 49% fewer missed cases.

6.3.2 Biomarker Significance Outcome

The combined feature importance analysis from RF and XGBoost confirms that Direct Bilirubin and Total Bilirubin are the most discriminative biomarkers for liver disease prediction in this dataset. This aligns with clinical hepatology: bilirubin accumulates when the liver cannot process it efficiently, making it the most direct biochemical proxy for hepatic dysfunction. The implication for clinical practice is that bilirubin measurements should always be prioritized in liver screening blood panels.

6.3.3 SMOTE Validation Outcome

The comparison of models trained with and without SMOTE confirms that class balancing is essential for this dataset. Without SMOTE, models consistently achieve higher accuracy (~73-75%) by defaulting to disease prediction (the majority class), but recall drops below 0.70. With SMOTE, recall improves to 0.73-0.76 range while maintaining competitive precision. The ROC-AUC improvement from ~0.82 to ~0.87 represents a clinically meaningful increase in discriminative ability.

6.3.4 Dashboard Usability Outcome

The 9-tab architecture successfully addresses multiple user personas: clinicians use Tabs 2, 3, and 4 for prediction and evaluation; data scientists use Tabs 5 and 6 for EDA and model analysis; and patients/health educators use Tabs 7, 8, and 9 for simulation and health guidance. The Lifestyle Simulator's real-time slider interface provides an intuitive demonstration of how biomarker threshold effects work — showing, for example, that SGPT values above ~145 U/L push risk probability above the 50% decision boundary.

6.4 Real-World Applicability

6.4.1 Primary Healthcare Screening

The most immediate real-world application is in primary healthcare centers and outpatient clinics in India. The five biomarkers required by the system (Total Bilirubin, Direct Bilirubin, SGPT, SGOT, ALP) are all included in standard liver function test (LFT) blood panels, which cost approximately Rs. 200-500 in Indian diagnostic labs. The dashboard could be integrated as a triage tool — after a patient's LFT results are available, a health worker could enter the values and receive a risk stratification that guides the decision to refer to a hepatologist.

6.4.2 Telemedicine and Rural Health

In rural India, where specialist healthcare access is severely limited, telemedicine platforms could integrate the prediction API to provide preliminary liver disease risk assessments. A patient's LFT results, obtainable from any basic diagnostic lab, could be submitted through a mobile interface. The risk stratification could inform remote physician consultations, enabling evidence-based triage without requiring specialist presence.

6.4.3 Population Health Surveillance

At a public health level, the system's demographic segmentation capability enables population-level analysis. Applying the model to aggregated LFT data from district health surveys could identify high-risk geographic clusters or demographic subgroups (e.g., senior male patients in industrial regions with elevated ALP), enabling targeted screening campaigns and resource allocation.

6.4.4 Medical Education

The EDA Visualization (Tab 5), Feature Relationships (Tab 6), and Future Risk Progression (Tab 8) tabs have immediate value as medical education tools. Students and residents can interactively explore how biomarker distributions differ between disease and non-disease populations, how feature importance rankings differ between RF and XGBoost, and at what exact biomarker thresholds disease risk crosses 50%. This interactive learning approach supplements textbook knowledge with data-driven insight.

6.5 Inference

The Indian Liver Patient Disease Predictor demonstrates that machine learning can deliver clinically meaningful liver disease risk stratification from standard blood test results. The project validates three key inferences:

(1) Ensemble methods substantially and consistently outperform individual classifiers for this dataset.

(2) Demographic segmentation reveals heterogeneous disease patterns that general models cannot capture.

(3) Interactive clinical dashboards can make complex ML predictions accessible and actionable for diverse user groups. The system is ready for pilot deployment and validation in a clinical setting.

CHAPTER 7: CONCLUSIONS AND RECOMMENDATIONS

7.1 Outline

This chapter synthesizes the key conclusions of the project, documents the limitations and constraints of the current system, and proposes directions for future enhancement.

The Indian Liver Patient Disease Predictor successfully delivers an end-to-end machine learning solution for liver disease prediction, achieving its stated objectives through rigorous methodology and comprehensive evaluation. The project demonstrates that a combination of SMOTE-balanced multi-model training, demographic segmentation, and Stacking Classifier ensemble can achieve 80% accuracy and 0.87 ROC-AUC on the ILPD dataset representing a meaningful advancement over baseline approaches.

7.2 Limitations and Constraints

- Dataset Size: The ILPD dataset contains only 583 records, limiting the robustness of demographic subgroup analysis (particularly for small groups such as Young Female patients). Findings may not generalize to larger or geographically diverse populations.
- Single Dataset: The system is trained exclusively on ILPD data from Andhra Pradesh. Biomarker reference ranges and disease prevalence can vary significantly across Indian regions, ethnic groups, and time periods.
- Missing Clinical Variables: Important clinical variables not captured in the ILPD dataset include: viral hepatitis serology (HBsAg, Anti-HCV), imaging findings, medication history (especially hepatotoxic drugs), alcohol consumption, BMI, and family history all of which significantly influence liver disease risk.
- SMOTE Limitations: SMOTE generates synthetic samples that may not correspond to clinically realistic patient profiles, potentially introducing subtle biases in model training. Borderline-SMOTE or ADASYN variants may produce more realistic synthetic samples.
- Model Interpretability: While feature importance scores are provided, the Stacking Classifier and ensemble models lack the case-level interpretability that clinicians require for individual patient decisions. SHAP (SHapley Additive exPlanations) values would provide case-level explanation.
- Prospective Validation: The system has not been validated on prospective clinical data. All evaluation is retrospective on the ILPD holdout set. Clinical utility cannot be confirmed without prospective or cross-dataset validation.
- Regulatory Compliance: The dashboard is currently a research and educational tool. Deployment as a clinical decision support system would require compliance with medical device regulations (e.g., CDSCO guidelines in India, FDA 510(k) in the US) and clinical trial validation.

7.3 Future Enhancements

7.3.1 Extended Dataset Integration

Future work should integrate additional liver disease datasets (e.g., Hepatitis C Dataset from UCI, NHANES liver data) for cross-dataset validation and transfer learning. Federated learning approaches could enable model training across multiple hospital systems without centralizing sensitive patient data.

7.3.2 Advanced Ensemble Techniques

Future research should explore: Bayesian optimization for hyperparameter tuning of the Stacking Classifier; Deep learning-based base learners (e.g., TabNet, neural network classifiers) within the stacking architecture; and AutoML frameworks (H2O.ai, Auto-sklearn) for automated model selection and hyperparameter optimization.

7.3.3 SHAP Explainability Integration

Adding a SHAP explanation tab (Tab 10) would provide case-level feature contribution waterfall charts, showing clinicians exactly which biomarker values drove a specific prediction. This is essential for clinical trust and regulatory compliance.

7.3.4 Natural Language Processing Integration

Integration of NLP models to parse unstructured clinical notes and extract liver disease risk factors could significantly expand the input feature space beyond structured blood test results.

7.3.5 Mobile Application Deployment

A progressive web application (PWA) or React Native mobile wrapper for the Streamlit dashboard would enable point-of-care use on smartphones and tablets, particularly valuable for rural healthcare settings.

7.3.6 Electronic Health Record Integration

API integration with hospital information systems (HIS) and laboratory information systems (LIS) would enable automated population of the prediction form with actual patient LFT results, eliminating manual data entry and associated errors.

7.3.7 Longitudinal Risk Tracking

The current system provides single-point predictions. A longitudinal tracking feature that maintains a patient's prediction history over multiple visits could enable early detection of deteriorating trends, even when individual values remain within 'borderline' ranges.

7.4 Inference

This project establishes a strong foundation for AI-assisted liver disease screening in the Indian healthcare context. The key conclusions are:

7. Ensemble methods (Random Forest, XGBoost, LightGBM, and especially Stacking Classifier) consistently and substantially outperform individual classifiers on the ILPD dataset, achieving 80% accuracy and 0.87 ROC-AUC.
8. SMOTE class balancing is essential for medical datasets with class imbalance, improving recall by up to 23 percentage points compared to unbalanced training.
9. Direct Bilirubin and Total Bilirubin are the most diagnostically important biomarkers in this dataset, supporting clinical guidelines that recommend bilirubin measurement as the primary liver function screening test.
10. The Stacking Classifier, trained on the Hybrid (age+gender segmented) dataset, is recommended for clinical screening use due to its optimal balance between false negatives (20) and false positives (12).
11. An interactive, multi-tab clinical dashboard makes complex ML predictions accessible and actionable for clinicians, patients, and health educators alike.

The Indian Liver Patient Disease Predictor represents a meaningful contribution to the application of computational data science to public health in India, and provides a strong foundation for continued research toward clinically validated, deployable AI-assisted hepatology screening.

REFERENCES

Journal References

1. Ramana, B. V., Babu, M. S. P., and Venkateswarlu, N. B. (2012). A critical comparative study of liver patients from USA and INDIA: An exploratory analysis. *International Journal of Computer Science Issues (IJCSI)*, 9(3), 506-516.
2. Singh, A., and Bhardwaj, M. (2018). Efficient feature selection and classification of liver disease using machine learning algorithms. *International Journal of Computer Science and Mobile Computing*, 7(4), 170-175.
3. Chawla, N. V., Bowyer, K. W., Hall, L. O., and Kegelmeyer, W. P. (2002). SMOTE: Synthetic minority over-sampling technique. *Journal of Artificial Intelligence Research*, 16, 321–357.
4. Chen, T., and Guestrin, C. (2016). XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 785–794.
5. Ke, G., Meng, Q., Finley, T., Wang, T., Chen, W., Ma, W., and Liu, T. Y. (2017). LightGBM: A highly efficient gradient boosting decision tree. In *Advances in Neural Information Processing Systems (NeurIPS)*, 30, 3146–3154.
6. Alam, M. Z., Rahman, M. S., and Rahman, M. S. (2020). A random forest based predictor for medical data classification using feature ranking. *Informatics in Medicine Unlocked*, 15, 100180.
7. Fernandez, A., Garcia, S., Herrera, F., and Chawla, N. V. (2018). SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary. *Journal of Artificial Intelligence Research*, 61, 863–905.
8. Obermeyer, Z., and Emanuel, E. J. (2016). Predicting the future — Big data, machine learning, and clinical medicine. *New England Journal of Medicine*, 375(13), 1216–1219.
9. Wolpert, D. H. (1992). Stacked generalization. *Neural Networks*, 5(2), 241–259.
10. Ma, L., Ma, C., Liu, Y., and Wang, X. (2020). Predicting the mortality of ICU patients with machine learning based on clinical variables. *IEEE Access*, 8, 17142–17148.
11. Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12, 2825–2830.
12. Waskom, M. L. (2021). Seaborn: Statistical data visualization. *Journal of Open Source Software*, 6(60), 3021.

Book References

1. Géron, A. (2022). *Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow*, 3rd ed. O'Reilly Media.
2. Hastie, T., Tibshirani, R., and Friedman, J. (2009). *The Elements of Statistical Learning*, 2nd ed. Springer.
3. He, H., and Ma, Y. (Eds.). (2013). *Imbalanced Learning: Foundations, Algorithms, and Applications*. Wiley-IEEE Press.

Website References

1. UCI Machine Learning Repository – Indian Liver Patient Dataset. Retrieved from: https://archive.ics.uci.edu/ml/datasets/ILPD+(Indian+Liver+Patient+Dataset)
2. Streamlit Documentation. Retrieved from: https://docs.streamlit.io/
3. World Health Organization – Liver Diseases Fact Sheet. Retrieved from: https://www.who.int/news-room/fact-sheets/detail/liver-cirrhosis
4. XGBoost Documentation. Retrieved from: https://xgboost.readthedocs.io/
5. imbalanced-learn Documentation – SMOTE. Retrieved from: https://imbalanced-learn.org/stable/references/generated/imblearn.over_sampling.SMOTE.html

APPENDIX A – DATASET SAMPLE

Table A.1 below presents the first 15 records of the Indian Liver Patient Dataset (ILPD) as used in this project, after target remapping (1=Disease, 0=No Disease).

Table : 12 - Sample Records from ILPD Table

Age	Gender	T-Bil	D-Bil	ALP	SGPT	SGOT	Total Prot.	Albumin	AGR	Label
65	F	0.7	0.1	187	16	18	6.8	3.3	0.90	1
62	M	10.9	5.5	699	64	100	7.5	3.2	0.74	1
62	M	7.3	4.1	490	60	68	7.0	3.3	0.89	1
58	M	1.0	0.4	182	14	20	6.8	3.4	1.00	1
72	M	3.9	2.0	195	27	59	7.3	2.4	0.40	1
17	M	0.9	0.3	202	22	19	7.4	4.1	1.20	0
25	F	0.9	0.2	154	16	12	7.0	3.5	1.00	1
29	F	0.9	0.3	202	14	11	6.7	3.6	1.10	1
64	M	0.9	0.3	310	61	58	7.0	3.4	0.90	0
74	F	1.1	0.4	214	22	30	8.1	4.1	1.00	1
38	M	1.8	0.8	342	168	441	7.6	4.4	1.30	1
33	M	1.6	0.5	165	15	23	7.3	3.5	0.92	0
40	F	0.9	0.3	293	232	245	6.8	3.1	0.80	1
51	M	2.2	1.0	610	17	28	7.3	2.6	0.55	1
63	M	0.9	0.2	194	52	45	6.0	3.9	1.85	0

APPENDIX B – SOURCE CODE OVERVIEW

The project source code is organized into two primary components: the Jupyter Notebook (ILPD_ML_2project.ipynb) for analysis and model development, and the Streamlit dashboard application (app2.py) for deployment.

B.1 Jupyter Notebook Structure

Table : 13 – Jupyter Notebook Structure Table

Step	Notebook Section	Description
1	Import Libraries	NumPy, Pandas, Matplotlib, Seaborn, Sklearn, XGBoost, LightGBM, imbalanced-learn
2	Load Dataset	pd.read_csv with shape verification and column listing
3	Data Preprocessing	Missing value imputation, label encoding, target remapping, feature scaling, SMOTE
4	EDA	Histograms, boxplots, correlation heatmap, summary statistics
5	Demographic Segmentation	Age groups (Young/Middle/Senior), Gender groups, Hybrid groups
6	Model Training Helpers	prepare_split(), get_models(), eval_model(), run_all_models() functions
7	Full Dataset Training	Train and evaluate all 6 models on SMOTE-balanced full dataset
7A/7B/7C	Subgroup Evaluation	Age-based, Gender-based, and Hybrid subgroup training and evaluation
8	Stacking Classifier	Define base models (RF, SVM, XGB), meta-model (LR), build and train StackingClassifier
9	Final Hybrid Comparison	Combined comparison of all 7 models on hybrid dataset with visualization
10	Final Evaluation	Best model identification per dataset type, full evaluation summary
11	Feature Importance	RF and XGBoost importance bar charts
12	ROC Curves	Multi-model ROC curve comparison plot
13	Model Saving	Export trained models for dashboard use

B.2 Dashboard Application Structure (app2.py)

Table : 14 - Dashboard Application Structure Table

Component	Implementation Detail
@st.cache_data	Caches dataset loading and preprocessing (runs once per session)

@st.cache_resource	Caches all 7 trained models (prevents retraining on user interactions)
st.tabs()	Creates 9-tab navigation structure
Tab 1 – Basic Info	st.metric(), st.dataframe(), custom feature reference table
Tab 2 – Overall Prediction	st.number_input() x5, st.button(), st.success()/st.error(), matplotlib gauge chart
Tab 3 – Advanced Prediction	st.selectbox() for type/model selection, dynamic input fields
Tab 4 – Evaluation Report	st.dataframe() with highlight_max() styling for metric tables
Tab 5 – EDA	matplotlib subplots for histograms and boxplots, seaborn heatmap
Tab 6 – Feature Relationships	RF/XGB horizontal barh charts, 6-panel confusion matrix grid, ROC curve overlay
Tab 7 – Lifestyle Simulator	st.slider() x5, live prediction on widget change, risk gauge chart
Tab 8 – Future Risk Progression	st.selectbox(), st.slider(), np.linspace() sweep, filled area line chart
Tab 9 – Diet & Prevention	st.expander() panels for food categories, st.warning() for clinical alert

Team Contribution :

- **Vani Varada – 24MIP10023**
  Handled age-based evaluation and machine learning model training.

- **Prathamesh Kumar – 24MIP10034**
  Worked on stacking-based model training, evaluated all models, managed data preprocessing and integration, and developed a unified web interface to present the final results.

- **Pushkar Singh – 24MIP10083**
  Conducted exploratory data analysis (EDA), implemented SMOTE for data balancing, and performed hybrid (age + gender) evaluation along with model training.

- **Ritesh Kumar Jha – 24MIP10088**
  Responsible for gender-based evaluation and machine learning model training.